Self-Exciting Order Flow

Hawkes processes: model, estimation, and how the branching ratio is misread

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The question

Orders cluster in time: one trade triggers others, cancellations come in bursts. **How much of market activity is triggered by market activity itself?**

- The share of endogenous events, the branching ratio, is a measure of reflexivity.
- Near-critical excitation links millisecond order flow to rough volatility over hours.
- For an LP, clustering means fills and adverse moves arrive in bursts, not evenly.

In this deck

1. The process and its cluster picture.
2. Exact likelihood and diagnostics.
3. The Hawkes price model and the signature plot.
4. Three ways an exponential fit misreads the branching ratio.
5. The criticality debate.

The process

For event types u, v (buy, sell, cancel, ...) with event times t_j^v, the intensity (s^{-1}) is

$$\Lambda_u(t) = \Lambda_u^0 + \sum_v \sum_{t_j^v < t} \phi_{uv}(t - t_j^v),$$

with baseline Λ_u^0 (s^{-1}) and kernels $\phi_{uv} \geq 0$ (s^{-1}). Write $G_{uv} = \int_0^\infty \phi_{uv}(t)\,\mathrm{d}t$ (dimensionless).

Cluster picture. Immigrants arrive at rate Λ^0; each type-v event has on average G_{uv} direct children of type u. The process is stationary iff the spectral radius $\rho(G) < 1$, and then

$$\bar{\Lambda} = (I - G)^{-1}\Lambda^0.$$

In one dimension $n = G$ is the **branching ratio**: the share of events that are endogenous.

Exponential kernel $\phi(t) = \alpha e^{-\beta t}$, with jump α and decay β (s^{-1}): $n = \alpha/\beta$, and the intensity is Markov, so the likelihood takes one pass.

Power-law kernel $\phi(t) \propto (t + \tau_0)^{-(1+\epsilon)}$, with cut-off τ_0 (s) and tail exponent $\epsilon > 0$: memory over many time scales, as in data.

Likelihood and diagnostics

On $[0, T]$ the log-likelihood of a point process with intensity Λ_u is

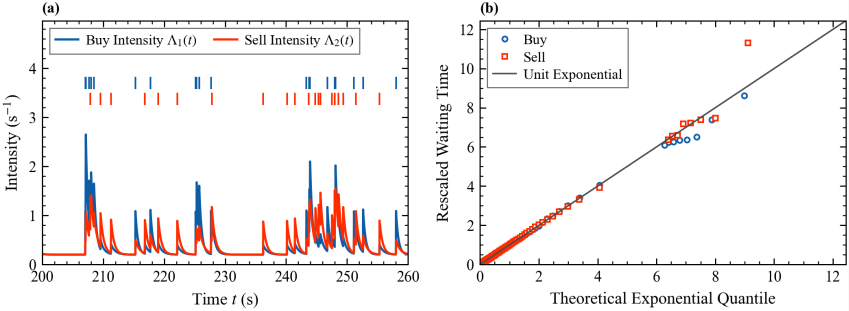
$$\ell = \sum_u \Big[\sum_j \ln \Lambda_u(t_j^u) - \int_0^T \Lambda_u(t)\,\mathrm{d}t \Big].$$

For exponential kernels both terms update recursively, so ℓ costs $O(N)$ for N events. Standard errors come from the inverse observed information.

Time-rescaling test. With compensator $\mathcal{A}_u(t) = \int_0^t \Lambda_u$, the increments $\mathcal{A}_u(t_j^u) - \mathcal{A}_u(t_{j-1}^u)$ are i.i.d. unit exponential under the true model. A Kolmogorov-Smirnov (KS) test checks them.

Nonparametric kernels. For slowly decaying kernels, Bacry et al. (2016) estimate ϕ from the empirical covariance of event counts by solving a Wiener-Hopf equation, over lags from microseconds to minutes.

Fitting buy and sell market orders



Bivariate exponential Hawkes model of buys and sells, $T = 10{,}000$ s: 3,972 buys and 4,523 sells, true spectral radius $\rho(G) = 0.518$. (a) Intensities over one minute.

MLE recovers all ten parameters within 2.23 standard errors. (b) Rescaled waiting times are unit exponential: KS $p = 0.94$ and 0.99.

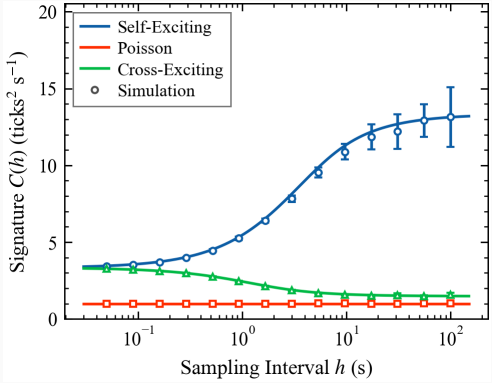
A Hawkes price: microstructure noise from cross-excitation

Up ticks N_+ and down ticks N_-, price $X = N_+ - N_-$ (ticks), self jump α_s, cross jump α_c, common decay β (Bacry et al., 2013). With $g = (\alpha_s - \alpha_c)/\beta$ and $\eta_H = \beta(1-g)$ (s^{-1}),

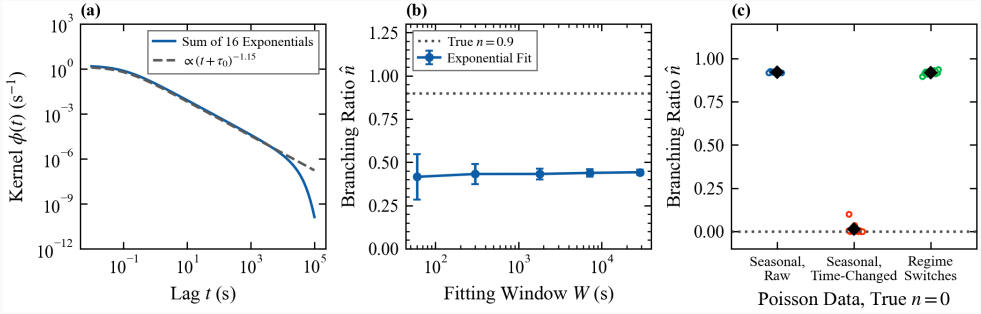
$$C(h) = 2\bar{\Lambda}\left[\frac{1}{(1-g)^2} + \left(1 - \frac{1}{(1-g)^2}\right)\frac{1-e^{-\eta_H h}}{\eta_H h}\right]$$

is the realized variance per unit time at sampling interval h ($\mathrm{ticks}^2\ \mathrm{s}^{-1}$), with $\bar{\Lambda}$ the mean rate per side.

- $\alpha_c > \alpha_s$: an up tick triggers down ticks, $C(h)$ falls with h: the signature plot of bid-ask bounce.
- Simulation within 3.1 s.e. over 42 points.



Three ways to misread the branching ratio



(a) Truth: a sum of 16 exponentials that follows $(t + \tau_0)^{-1.15}$ from 0.1 s to about 10^4 s, with $n = 0.9$.

(b) An exponential kernel fitted on windows from 1 min to 8 h finds $\hat{n} = 0.42 \pm 0.07$ to 0.44 ± 0.01, with decay time $1/\hat{\beta} \approx 0.30$ s. It misses more than half the excitation at every window length.

(c) Pure Poisson data ($n = 0$), 12 days each: a U-shaped intraday rate gives $\hat{n} = 0.919 \pm 0.001$; after a time change by the estimated profile, 0.014 ± 0.008. Random rate regimes: 0.919 ± 0.003.

Why the fits go wrong

Nonstationary baseline. A slow change in the baseline rate looks to the likelihood like a long cluster. A slowly decaying exponential ($1/\hat{\beta}$ of about 100 s here) absorbs it and reports $\hat{n}$ close to one. This is the result of Filimonov and Sornette (2015): Poisson data with regime changes give spurious near-critical branching ratios.

Remedy. Estimate the seasonal profile across days and change time by its integral, $t \mapsto \int_0^t \hat{\Lambda}^0(s)\,\mathrm{d}s$, before fitting. Regime changes need a model of their own.

Misspecified kernel. A single exponential has one time scale. Faced with excitation spread over six decades, the likelihood settles on the fast part and reports less than half of n, whatever the window.

Remedy. Use kernels with many time scales (sums of exponentials or power laws), nonparametric estimation, and check residuals with the time-rescaling test.

Other traps. Coarse timestamps and simultaneous events bias short-lag excitation; edge effects (no history before the window) bias $\hat{n}$ down.

What the data show

- **Critical reflexivity.** For mid-price changes of the E-mini S&P 500 futures, 1998 to 2011, a power-law kernel with exponent near -1.15 below 10^3 s and near -1.45 from 10^3 to 10^6 s integrates to about one in every two-month period: the market is, and was, close to critical (Hardiman et al., 2013).
- **Why exponential fits disagreed.** Fitting exponential kernels on 30-minute windows to synthetic critical power-law data gives sub-critical branching ratios that rise as the short-time cut-off falls, which can mimic a rise in reflexivity over the years (Hardiman et al., 2013).
- **The counterpoint.** Regime shifts and outliers bias $\hat{n}$ upward, and calibration on Poisson data with regime changes gives $n \approx 1$ when the truth is zero (Filimonov and Sornette, 2015).
- **Many dimensions.** Nonparametric estimation on an eight-dimensional order book model finds slowly decaying kernels over lags from about 10 μs to 100 s (Bacry et al., 2016). Survey: Bacry et al. (2015).

From order flow to rough volatility

- As $\rho(G) \to 1$ with the horizon, suitably rescaled nearly unstable Hawkes processes converge to an integrated Cox-Ingersoll-Ross-type process (Jaisson and Rosenbaum, 2015). Later work with heavy-tailed kernels obtains **rough** limits, with Hurst exponent below one half.
- Adding no-arbitrage, buy-sell asymmetry and metaorder splitting gives the rough Heston model with leverage (El Euch et al., 2018); recent work adds common price and volatility jumps (Wang et al., 2026).
- For an LP: the branching ratio estimated from milliseconds of order flow and the volatility hedged over hours are two views of one process. Mis-estimating n (previous slides) mis-estimates both.

Takeaways

Fit with kernels that have many time scales; remove seasonality first; test residuals; and distrust a branching ratio near one until regime changes are ruled out. The criticality question is open because each answer depends on these choices.

Reproduce

Code: `hawkes.py`. Experiments: `e4_hawkes.py`, `e5_signature.py`, `e17_hawkes_pitfalls.py`.

```
from stochastic_lob import hawkes as hk
w, beta = hk.power_law_weights(n=0.9, eps=0.15, tau0=0.1, beta_min=1e-4)
model = hk.SumExpHawkes1D(mu=0.05, w=w, beta=beta)
t = model.simulate(T=3e4, rng=1)
mu, n_hat, beta_hat, se = hk.fit_exp_hawkes_1d(t, 3e4)
```

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